DSA Roadmap - 2024

A **Three Months** Data Structure and Algorithms Roadmap for all Levels of preparations.

Get Placements Ready in 5 Months with DSA Prime - With Doubts Support 💬 Coding Contests 🥷 and Mock Test Series 🎯

Preparation Level

📅 Everyday for **3 Months**

	Daily Min Hours	Daily Min Score	Ready
Basic	1 Hr	200	70%
Serious	3 Hr	400	95%
Insane	5 Hr	600	99%

Pre-requisites:

- Working knowledge of Any Object-oriented Programming Language. Preferably C++, Java or Python.

Month 1: Foundation

Pre-requisites

- Have a basic understanding of Arrays, Linked Lists, Stack and Queues

Problems

Rating	Topic	Easy (100)	Medium (200)	Hard (400)	Patterns
5 ⭐	Arrays	10	6	3	1. Two Pointers Algorithm 2. Sorting Based Problems 3. Kadane's Algorithm 4. Array Rotations 5. Subarrays Based Problems
4 ⭐	Linked List	8	4	2	1. Nodes Manipulation 2. Reversing, Reordering List 3. Manipulating Two Lists
4 ⭐	Sorting	3	3	1	1. Merge Sort 2. Quick Sort 3. Inversion count
3 ⭐	Stacks	8	3	2	1. Implementation using Stack 2. Infix, Prefix, Postfix 3. Previous smaller element 4. Window-based problems
3 ⭐	Queues	6	3	1	1. Queue using Stack and vice-versa 2. Sliding window-based problems
3 ⭐	Backtracking	3	4	2	1. Knights' Tour 2. Rat in a Maze 3. N-Queen Problem
2 ⭐	Recursion	6	3	2	1. nCr, Josephus 2. Generating Palindrome Strings, Subsets of String

Additional

- Keep preparing Notes and breakdown of important Problems and concepts.
- Beginners should focus more on the High Rated Topics.

Month 2: Important Concepts

Pre-requisites

- Revise the Notes and important concepts from the previous month
- Have a basic understanding of Hashing, Binary Searching and Binary Trees

Problems

Rating	Topic	Easy (100)	Medium (200)	Hard (400)	Patterns
5 ⭐	Hashing	10	6	2	1. Substring, subarray based Hashing 2. String and Matrix-based Hashing Problems
4 ⭐	Binary Searching	6	3	2	1. Sorted and rotated arrays 2. square root, median of array 3. Using a middle point to solve the array problems like the Tree Cutter Problem
4 ⭐	Binary Tree	8	4	2	1. DFS Traversal of Binary Tree 2. Problems using BFS Traversal of Binary Tree 3. Upward Traversal based Problems.
4 ⭐	Binary Search Trees	4	5	2	1. Implementation of BST 2. Sorted Array Problem in BST, like Two-sum or Minimum Absolute Difference
2 ⭐	Priority Queue	3	4	1	1. Heap Implementations 2. Merge k-sorted Lists 3. Median or running stream
2 ⭐	Bit Manipulation	6	2	1	1. Bit masking 2. Problems using XOR properties
2 ⭐	Mathematics	6	4	1	1. problems using Euclid GCD 2. Modulo Arithmetics, Catalan's Numbers, Fermat's Little Theorem, Pigeonhole principle

Additional

- Keep preparing Notes and breakdown of important Problems and concepts.
- Solve more Medium level problems and don’t spend too much time on one problem.
- Start participating in mock interviews with your friends, Do pair programming.

Month 3: Advanced Topics and Mock Tests

Pre-requisites

- Revise the Notes and important concepts from the previous months.
- Have a basic understanding of Dynamic Programming and Graphs

Problems

Rating	Topic	Easy (100)	Medium (200)	Hard (400)	Patterns
5 ⭐	Mock Tests	2	4	2	1. Practice company wise Mock Tests 2. Participate in Coding contests.
4 ⭐	Greedy Programming	3	5	2	1. Candy Distribution based Problems 2. Activity Selection based Problems
4 ⭐	Dynamic Programming	10	8	4	1. Linear DP 2. LCS Based DP 3. LIS Based DP 4. 0/1 Knapsack based DP 5. Grid-based DP 6. Partitioning DP